

# Micah Shields

Email: [micah@micah.es](mailto:micah@micah.es) | LinkedIn: <https://www.linkedin.com/in/micahshields>  
Portfolio: <https://micah.es> | Location: Spain | Right to Work: EU & UK (Unlimited)

## SUMMARY

Backend Software Engineer with nearly 4 years of experience building and optimising AI-powered production systems using Python, FastAPI, PostgreSQL, and Vertex AI. Previously worked as a Database Administrator, giving me a strong foundation in databases, performance optimisation, and scalable data systems.

## EXPERIENCE

### Professional Development / Upskilling | February 2026 – Present

Decided to have a period of professional development which will be dedicated to:

- Purchasing my own domain and website: micah.es and programming a portfolio to host on it.
- Completing a Data Structures and Algorithms course and sharpening my problem-solving skills through LeetCode.
- Completing a Systems Design course and implementing projects using the knowledge gained from it.
- Completing a SOLID / clean code course.
- Improving my German and Portuguese.

### Mid Software Engineer | StudyHall AI | July 2022 – February 2026 | Remote (UK-based company)

- Achieved significant performance gains on the VertexAI-based EPUB-processing pipeline by implementing Redis caching, fixing race conditions, optimising DB connections, optimising SQL queries, migrating from Pandas to Polars, and eliminating redundant data loads
- Optimised and maintained independently the core PostgreSQL database, cutting heavy query runtimes by **~90%**
- Key contributor of optimisation and implementation of critical FastAPI-powered data science services (planning, exam & module creation, assessment, reporting) used by thousands of users
- Engineered robust Python pipelines to ingest fundamental educational datasets used by the StudyHall AI platform. Replacing obsolete VBA scripts, therefore reducing time to ingest content by **~90%**
- Optimised and maintained a TypeScript-based Postgres → MongoDB seeder used by the client team heavily reducing runtime by identifying and refactoring N+1 queries
- Built and maintained internal tools with Retool and Streamlit, streamlining workflows and cutting manual processes
- Assisted and guided junior engineers on best practices, code quality, and system design, raising overall team delivery and reliability

### English Teacher (TEFL) | Your Language Club Getafe | October 2021 – July 2022 | Madrid, Spain

- Delivered over **400** English lessons, building deep communication skills and the ability to explain complex concepts clearly – an asset in technical collaboration and documentation

### Database Administrator | Ministry of Furniture | October 2017 – October 2021 | Neath, UK

- Maintained and optimised independently two product databases (MySQL & Microsoft SQL Server) with 6,000+ SKUs and 300,000+ variants, reducing annual update time by **~80%** through SQL and VBA automation

## SKILLS

- **Computer Languages:** Python, SQL, PL/SQL, TypeScript, JavaScript, VBA, HTML, CSS, Neo4j Cypher
- **Frameworks:** FastAPI, Pydantic, Numpy, Pandas, SQLAlchemy, SQLAlchemyModel, BeautifulSoup, Django, Astro, Tailwind
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Neo4j
- **Tools:** GitHub, Docker, Google Cloud, Sentry, Retool, Streamlit, JIRA
- **Natural Languages:** English (Native), Spanish (C2), Portuguese (A2), German (A1)
- **Certifications:** DELE C2 Spanish, TEFL Distinction